

BIO-DATA

1. Name: **Arijit Mondal**
2. Designation: Assistant Professor Grade I
3. Department: Dept of CSE
4. Institution: IIT Patna
5. Contact Number: 9632127613
6. Email: arijit@iitp.ac.in
7. Date of Birth: 28.12.1979
8. Gender: Male
9. Academic Qualification: (Undergraduate Onwards)

S.No.	Degree	Year	Subject	University/Institution	% of Marks
1	B.Tech	2001	EE	IIT Kharagpur	8.40
2	M.S.	2004	CSE	IIT Kharagpur	9.10
3	Ph.D.	2010	CSE	IIT Kharagpur	-

10. Ph.D. Thesis Title: A Symbolic Event Propagation Approach for Solving Timing Problems of Digital Circuits

Guide Name : Prof. P. P. Chakrabarti & Prof. Pallab Dasgupta

Institution : IIT Kharagpur

Year of Award : 2010

11. Work Experience (in chronological order)

S.No.	Position held	Institution	From	To
1	Senior Member of Technical Staff	Berkeley Design Automation	April, 2009	February, 2012
2	Member of Technical Staff	Cadence Design System	March, 2012	October, 2012
3.	Assistant Professor	IIT Patna	November, 2012	December, 2019
4.	Assistant Professor	CoEAI, IIT Kharagpur	December, 2019	July 2021
5.	Assistant Professor	IIT Patna	August 2021	Till now

12. Selected Publications (List of papers published in SCI Journals, in year wise descending order).

1. Adity Ghosh, Arnab Sarkar, Arijit Mondal, Dynamic Scheduling of Aperiodic Real-time Task Graphs on Heterogeneous Edge Computing Environments, ACM TECS 2025 (Accepted for publication)
2. Suraj Meshram, Arnab Sarkar, Arijit Mondal, A Tunable Generic Meta-Heuristic Framework for Balancing Assembly Line Systems in Manufacturing, To appear in ACM TECS, 2025
3. Dipankar Mandal, Arnab Sarkar, Arijit Mondal, A Discrete Partial Charging enabled Dynamic Programming Strategy for Optimal Fixed-Route Electric Vehicle Charging, To appear in ACM TECS, 2025
4. Ashiqur Rahaman Molla, Ram Mohan Chowdary Kota, Jaishree Mayank, Arnab Sarkar, Arijit Mondal, Soumyajit Dey, Minimizing Backbone Ethernet Traffic for Enabling Inter-zonal Messages in Software-Defined Vehicles, To appear in IEEE Embedded Systems Letters, 2025
5. Suraj Meshram, Sanket Jaipurkar, Arnab Sarkar, Arijit Mondal, An Efficient Iterative Beam Search for Human-Robot Collaborative Assembly Line Balancing, To appear in IEEE Embedded Systems Letters, 2025
6. Jyoti Kumari, Arijit Mondal, Jimson Mathew, Fourier-driven Lightweight Token Mixing Model for Efficient Time Series Forecasting, To appear in IEEE Transactions on Artificial Intelligence, 2025
7. Kavitha Yogaraj, Brian Quanz, Tarrun Vikas, Arijit Mondal, Samrat Mondal, Post-variational classical quantum transfer learning for binary classification, *Sci Rep* **15**, 23682 (2025). <https://doi.org/10.1038/>
8. Ranjit Kumar Mishra, Arijit Mondal, Jimson Mathew, Nystromformer based cross-modality transformer for visible-infrared person re-identification, Scientific Reports, Nature Publishing Group, May, 2025, Vol-15, Issue-1
9. Jaishree Mayank, Arijit Mondal, A Unified Job Scheduler for Optimization of Different System Performance Metrics, Concurrency and Computation: Practice and Experience, Wiley (accepted)
10. Anamika Dey, Somrita Sarkar, Arijit Mondal, and Pabitra Mitra. "Spatio-Temporal NDVI Prediction for Rice Crop." *SN Computer Science* 6, no. 3 (2025): 1-13.
11. Ranjit Kumar Mishra, Arijit Mondal, Jimson Mathew, Swin Transformer with Part-Level Tokenization for Occluded Person Re-identification, *Machine Vision and Applications* 36.1 (2025): 1-17.
12. Shivesh Kumar Sharan, Surajit Kumar Paul, Jyoti Kumari, and Arijit Mondal. "Artificial Neural Network-Based Prediction of Complete Forming Limit Curves for Steel in Sheet Metal Forming." *Journal of Alloys and Metallurgical Systems* (2025): 100166.
13. Shubhajit Dutta, Manaar Alam, Arijit Mondal, Debdeep Mukhopadhyay, Partha P Chakrabarti, Decision Guided Robust DL Classification of Adversarial Images Combining Weaker Defenses, IEEE Journal on Emerging and Selected Topics in Circuits and Systems. 2024
14. Somrita Sarkar, Anamika Dey, Ritam Pradhan, Upendra Mohan Sarkar, Chandranath Chatterjee, Arijit Mondal, and Pabitra Mitra, "Crop Yield Prediction Using Multimodal Meta-Transformer and Temporal Graph Neural Networks", Accepted for publication in IEEE Transactions on AgriFood Electronics, 2024
15. Suraj Singh, Ashiqur Rahaman Molla, Arijit Mondal, Soumyajit Dey, "Co-designing Perception-based Autonomous Systems on CPU-GPU platforms", *IEEE Embedded Systems Letters* (Accepted), 2024
16. Shruti Verma and Jyoti Kumari and Arijit Mondal and T. Santhanakrishnan and Jimson Mathew and Sumanta Gupta, Modeling and simulation of Φ -OTDR system suitable for underwater intrusion detection and classification, Optik, Elsevier, Vol-311, September 2024.
17. Surbhi Raj, Jimson Mathew, Arijit Mondal, Generalized and Robust Model for GAN-generated Image Detection, Pattern Recognition Letters, Elsevier, Volume 182, June 2024
18. Ritam Sarkar, Pranay Saha, Samrat Mondal, Arijit Mondal, MAGE: Microgrids with Advanced Grid Efficiency through Battery-Aware EV Management, Elsevier Sustainable Cities and Society, Vol 107, July, 2024
19. Sanghamitra Mishra, Arijit Mondal, Smart Mondal, Charging Station Siting and Sizing Considering Uncertainty in Electric Vehicle Charging Demand Distribution, IEEE Transactions of Intelligent Transportation Systems (2024 accepted for publication)

20. Pranay Saha, Nilotpal Chakraborty, Arijit Mondal, Samrat Mondal, Intelligent Real-time Utilization of Hybrid Energy Resources for Cost Optimization in Smart Microgrids, IEEE Systems journal, vol. 18, no. 1, pp. 186-197, March 2024
21. Jyoti Kumari, Jimson Mathew, Arijit Mondal, MAD-MEL: Combining Entity and Metric Learning for Anomaly Detection in Multivariate Time Series, IEEE Sensors journal, vol. 24, no. 3, pp. 3144-3156, 1 Feb.1, 2024
22. Surbhi Raj and Jimson Mathew and Arijit Mondal, FDT: A python toolkit for fake image and video detection, SoftwareX, Volume 22, May 2023, 101395
23. Fazail Amin, Arijit Mondal, Jimson Mathew, Person Re-Identification Using Selective Transformation Learning, Multimedia Tools and Applications, Springer, 2023.
24. Sangahmitra Mishra, Arijit Mondal, Samrat Mondal, A Multi-Objective Optimization Framework for Electric Vehicle Charge Scheduling with Adaptable Charging Ports, in IEEE Transactions on Vehicular Technology, doi: 10.1109/TVT.2022.3231901
25. Pranay Kumar Saha, Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Optimal Sizing and Efficient Routing of Electric Vehicles for a Vehicle-On-Demand System," IEEE Transactions on Industrial Informatics, vol. 18, no. 3, pp. 1489-1499, March 2022, doi: 10.1109/TII.2021.3091597
26. Fazail Amin, Arijit Mondal, and Jimson Mathew, "A Large Dataset With a New Framework for Abandoned Object Detection in Complex Scenarios," IEEE MultiMedia, May 2021
27. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Intelligent Charge Scheduling and Eco-Routing Mechanism for Electric Vehicles: A Multi-Objective Heuristic Approach," Elsevier Sustainable Cities and Society, Vol 69, June 2021, 102820
28. Pranay Kumar Saha and Arijit Mondal, "Efficient Online Heuristic Approach for Handling Fluctuation in Renewable Energy in a Microgrid," IEEE Systems Journal, vol. 16, no. 2, pp. 2471 - 2482, June 2022
29. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Efficient Load Control Based Demand Side Management Schemes Towards a Smart Energy Grid System," Elsevier Sustainable Cities and Society, vol. 59, 102175, August 2020.
30. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Scheduling Interdependent Smart Appliances with Mixed-Preemption Policy for Peak Load Minimization," Taylor and Francis IETE Journal of Research, 66:6, 797-805, December 2020.
31. Basina D. R., Kumar S. , Padhi S. , Sarkar A. , Mondal A. , Kirthi R., "Brownout Based Blackout Avoidance Strategies in Smart Grids", IEEE Transactions on Sustainable Computing, vol. 6, no. 4, pp. 586-598, 1 Oct.-Dec. 2021, doi: 10.1109/TSUSC.2020.3014077.
32. Jaishree Mayank, and Arijit Mondal, "Efficient SAT Encoding Scheme for Schedulability Analysis of Non-preemptive Tasks on Multiple Computational Resources." Journal of Systems Architecture, June 2020.
33. Jana R. L., Dey S. , Dasgupta P. , Mondal A. "Automated Planning for finding Alternative Bug Traces", IET Computers & Digital Techniques, 14(6), 322-335, 2020.
34. Niraj Kumar, Jaishree Mayank and Arijit Mondal, "Reliability aware Energy Optimized Scheduling of Non-preemptive Periodic Real-Time Tasks on Heterogeneous Multiprocessor System," IEEE Transactions on Parallel and Distributed Systems, vol. 31, no. 4, pp. 871-885, 1 April 2020, doi: 10.1109/TPDS.2019.2950251.
35. Jaishree Mayank and Arijit Mondal, "Polynomial time schedulability test for periodic non-preemptive 2-task system," Information Processing Letters: Volume 154, February 2020.
36. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Multi-Objective Optimal Scheduling Framework for HVAC Devices in Energy Efficient Buildings," IEEE Systems Journal, vol. 13, no. 4, pp. 4398-4409, Dec. 2019
37. Jaishree Mayank, Arijit Mondal and Arnab Sarkar, "Control-schedule co-design for fast stabilization in real time systems facing repeated reconfigurations," Design Automation for Embedded Systems, Springer, 2019
38. Jaishree Mayank and Arijit Mondal, "An Integer Linear Programming Framework for Energy Optimization of Non-Preemptive Real-Time Tasks on Multiprocessors," Journal of Low Power Electronics, vol. 15, no. 2, pp: 162–167, June 2019.

39. Niraj Kumar, Arijit Mondal, "Timing Analysis of Precedence Constraint Messages Scheduled over Dynamic Segment of FlexRay," IEEE Transactions on Automation Science and Engineering, vol. 17, no. 1, pp. 222-236, Jan. 2020, doi: 10.1109/TASE.2019.2907959
40. Ananya Singla, Varsha Agarwal, Sudip Roy and Arijit Mondal, "Reliability Analysis of Mixture Preparation using Digital Microfluidic Biochips," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 38, no. 4, pp. 654-667, April 2019.
41. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Efficient Scheduling of Non-Preemptive Appliances for Peak Load Optimization in Smart Grid," in IEEE Transactions on Industrial Informatics, vol. 14, no. 8, pp. 3447-3458, Aug. 2018.
42. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Intelligent Scheduling of Thermostatic Devices for Efficient Energy Management in Smart Grid," in IEEE Transactions on Industrial Informatics, vol. 13, no. 6, pp. 2899-2910, Dec. 2017.
43. Arijit Mondal, P. P. Chakrabarti, Pallab Dasgupta, "Symbolic Event Propagation Based Minimal Test Set Generation for Robust Path Delay Faults", ACM Transactions on Design Automation of Electronic Systems, Volume 17 Issue 4, October 2012.
44. Arijit Mondal, P. P. Chakrabarti, Pallab Dasgupta, "Statistical static timing analysis using symbolic event propagation", IET Circuit, Device and Systems, Vol 1, No 4, pages 283-291, 2007.
45. Arijit Mondal, P. P. Chakrabarti, "Reasoning about timing behavior of digital using symbolic event propagation and temporal logic", IEEE Transaction on Computer Aided Design of Integrated Circuits and Systems, Vol 25, No. 9, pages 1793-1814, 2006.

13. List of Projects implemented

Projects	Funding agency	Status	Budget, Role
Smart Energy Management in Office Building	IIT Patna	Completed	12L, PI
A Software Tool for the Planning and Design of Smart Micro Power Grids	IMPRINT-I	Completed	202L, Co-PI
Special Manpower Development Program - Chip to System Design (SMDP-C2SD)	MeitY	Completed	100L, Co-PI
Design and FPGA prototyping of multicarrier multiple access schemes for variable rate multimedia satellite communication	MeitY	Completed	90L, Co-PI
Development of Planning and Designing Tool for Smartly Adopting Electric Vehicles in Indian Cities	IMPRINT-II	Completed	63L, Co-PI
V2D: Video-to-Description Generation using Deep Learning	DST	Completed	32L, PI
Deep Learned Detection and Classification of Multiple Intrusions Using WDM Intensity and Phase-Sensitive OTDR in Underwater Environment	NRB, DRDO	Completed	46L, PI
A Smart revenue optimization framework for on-demand scheduling & routing of heterogeneous EVs	TIH, IIT Bhilai	Ongoing	75L, PI
AI for Physical Marketplace Management	RAKSHAK, DST	Completed	10L, PI
AI Based Digital Twin Support System for Mobility Analytics	IIT Kharagpur	Completed	25L, PI
Campus Rakshak: An Agent-based Simulation Framework for Assessment of Intervention Strategy against Covid-19 Spreading in Campuses	RAKSHAK, DST	Completed	10L, Co-PI

AI for Agriculture and Food Sustainability	DST	Completed	131L, Co-PI
AWS:NAIRP:Creating a Collateral for Supporting Popular ML Applications on AWS Infrastructure through National AI Resource Platform	Amazon	Completed	90L, Co-PI
IIT Patna Centre of Excellence in Cyber Crime Prevention against Women and Children -AI-based Tools for Women and Child Safety	BPRD, MHA	Completed	283L, Co-PI
Deep Learned Detecation and Classification of Multiple Intrusions Using WDM Intensity and Phase-Sensitive OTDR in Underwater Environment-Phase II	NRB, DRDO	Ongoing	85L, Co-PI

SURAJ KUMAR

Contact Address

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Email: suraj.kumar@inria.fr

Interests

Low-rank approximations, Deep neural networks, Matrix and tensor computations, Parallel algorithms

Professional Experience

- Inria Lyon** Lyon, France
ISFP (Junior Researcher) Oct 2022 – present
- Inria Paris** Paris, France
Postdoctoral Researcher Nov 2019 – Sep 2022
- Pacific Northwest National Laboratory** Richland, Washington, USA
Postdoctoral Research Associate May 2018 – Oct 2019
- Ericsson Research** Bangalore, India
Senior Engineer Aug 2017 – Feb 2018
- IBM India Research Lab** New Delhi, India
Software Engineer Jul 2012 – Nov 2013

Education

- Inria Bordeaux, University of Bordeaux** Bordeaux, France
Doctorate of Philosophy Dec 2013 – Apr 2017
- Indian Institute of Science** Bangalore, India
Master of Engineering in Computer Science and Engineering Aug 2010 – Jun 2012

Publications

The papers with * have author names in alphabetical order.

- *Brief Announcement: Minimizing Communication for Parallel Symmetric Tensor Times Same Vector Computation
Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse, Mathieu V  rit  
ACM Symposium on Parallelism in Algorithms and Architectures (*SPAA 2025*), Jul 2025, Portland, Oregon, USA.
- *Communication Lower Bounds and Optimal Algorithms for Symmetric Matrix Computations
Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse, Mathieu Verite
ACM Transactions on Parallel Computing (*TOPC*), 2025, Volume 12, Issue 2.
- *Communication Lower Bounds and Optimal Algorithms for Multiple Tensor-Times-Matrix Computation
Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse
SIAM Journal on Matrix Analysis and Applications (*SIMAX*), 2024, 45(1).

4. *Parallel Memory-Independent Communication Bounds for SYRK
Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse
ACM Symposium on Parallelism in Algorithms and Architectures (*SPAA 2023*), Jun 2023, Orlando, FL, USA.
5. *Brief Announcement: Tight Memory-Independent Parallel Matrix Multiplication Communication Lower Bounds
Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse
ACM Symposium on Parallelism in Algorithms and Architectures (*SPAA 2022*), Jul 2022, Philadelphia, PA, USA.
6. NWChemEx – computational chemistry for the exascale era
Karol Kowalski, Edoardo Aprà, Raymond Bair, Jeffery S. Boschen, Eric J. Bylaska, Jeff Daily, Wibe A. de Jong, Thom Dunning, Niranjana Govind, Robert J. Harrison, Kristopher Keipert, Sriram Krishnamoorthy, Suraj Kumar, Erdal Mutlu, Bruce Palmer, Ajay Panyala, Bo Peng, Ryan M. Richard, T. P. Straatsma, Edward F. Valeev, Marat Valiev, Hubertus J. J. van Dam, David B. Williams-Young, Chao Yang, Marcin Zalewski, Theresa L. Windus
Chemical Reviews 2021, Volume 121(8), 4962-4998.
7. *Analysis of a List Scheduling Algorithm for Task Graphs on Two Types of Resources
Lionel Eyraud-Dubois, Suraj Kumar
IEEE International Parallel & Distributed Processing Symposium (*IPDPS 2020*), May 2020, New Orleans (Virtual), Louisiana, USA.
8. Performance Models for Data Transfers: A Case Study with Computational Chemistry Kernels
Suraj Kumar, Lionel Eyraud-Dubois, Sriram Krishnamoorthy
International Conference on Parallel Processing (*ICPP 2019*), Aug 2019, Kyoto, Japan.
9. *Fast Approximation Algorithms for Task-Based Runtime Systems
Olivier Beaumont, Lionel Eyraud-Dubois, Suraj Kumar
Concurrency and Computation: Practice and Experience (*CCPE*), Wiley, 2018, 30:e4502.
10. *Approximation Proofs of a Fast and Efficient List Scheduling Algorithm for Task-Based Runtime Systems on Multicores and GPUs
Olivier Beaumont, Lionel Eyraud-Dubois, Suraj Kumar
IEEE International Parallel & Distributed Processing Symposium (*IPDPS 2017*), May 2017, Orlando, Florida, USA.
11. *Scheduling of Linear Algebra Kernels on Multiple Heterogeneous Resources
Olivier Beaumont, Terry Cojean, Lionel Eyraud-Dubois, Abdou Guermouche, Suraj Kumar
International Conference on High Performance Computing, Data, and Analytics (*HiPC 2016*), Dec 2016, Hyderabad, India.
12. *Are Static Schedules so Bad? A Case Study on Cholesky Factorization
Emmanuel Agullo, Olivier Beaumont, Lionel Eyraud-Dubois, Suraj Kumar
IEEE International Parallel & Distributed Processing Symposium (*IPDPS 2016*), May 2016, Chicago, IL, USA. IEEE, 2016.
13. *Bridging the Gap between Performance and Bounds of Cholesky Factorization on Heterogeneous Platforms
Emmanuel Agullo, Olivier Beaumont, Lionel Eyraud-Dubois, Julien Herrmann, Suraj Kumar, Loris Marchal, Samuel Thibault
Heterogeneity in Computing Workshop (*HCW 2015*), IPDPS 2015, Hyderabad, India.

14. Performance Optimizations for TTI RTM on GPU based Hybrid Architectures
Ankur Narang, Suraj Kumar, Ananda S. Das, Michael Perrone, David Wade, Kristian Bendiksen, Vidar Slatten, Tor Erik Rabben
10th Biennial International Conference & Exposition, 2013.
15. Maximizing TTI RTM Throughput for CPU+GPU
Ankur Narang, Suraj Kumar, Jyothish Soman, Michael Perrone, David Wade, Kristian Bendiksen, Vidar Slatten, Tor Erik Rabben
75th EAGE Conference & Exhibition incorporating SPE EUROPEC 2013, London, UK.
16. Optimized Association Rule Mining using Genetic Algorithm
M Anandhavalli, Suraj Kumar, Sudhanshu, Ayush Kumar
Bioinfo Publications, Advances in information Mining. ISSN: 0975-3265, Volume 1, Issue 2, 2009.

Posters

- Scheduling of Cholesky Factorization with Lookahead Information
Suraj Kumar, HiPC 2016.
- Scheduling Strategies and Bounds for Cholesky Factorization on Heterogeneous Platforms
Suraj Kumar, SC 2016.
- Scheduling of Task-Based Linear Algebra Kernels on Heterogeneous Resources
Suraj Kumar, IPDPS 2016 PhD Forum.

Teaching

- Master2 course: Resource-aware computations on CPUs and GPUs (Fall 2025)
- Master2 course: Data-aware algorithms for matrix and tensor computations (Fall 2023, Fall 2024)

Awards, Competitions and Miscellaneous Information

- Student travel awards to attend SC 2016 and IPDPS 2016.
- All-India Rank 28 (top 0.03%) at the Gate Examination 2010, out of a total of about 107,000 candidates.
- Secured 1st position in programming contest organized at IIT Guwahati, India in their technical fest TECHNICHE09.

Professional Services

- Reviewer for the following conferences: SC 2025, SC 2024, ICPP 2022
- External reviewer for the following conferences: SPAA 2023, ICPP 2019
- Reviewer for the following international journals: IJPP since March 2018, CALC since March 2020, SIMAX since May 2020, TOMS since May 2020, SISC since May 2021, TPDS since May 2022.